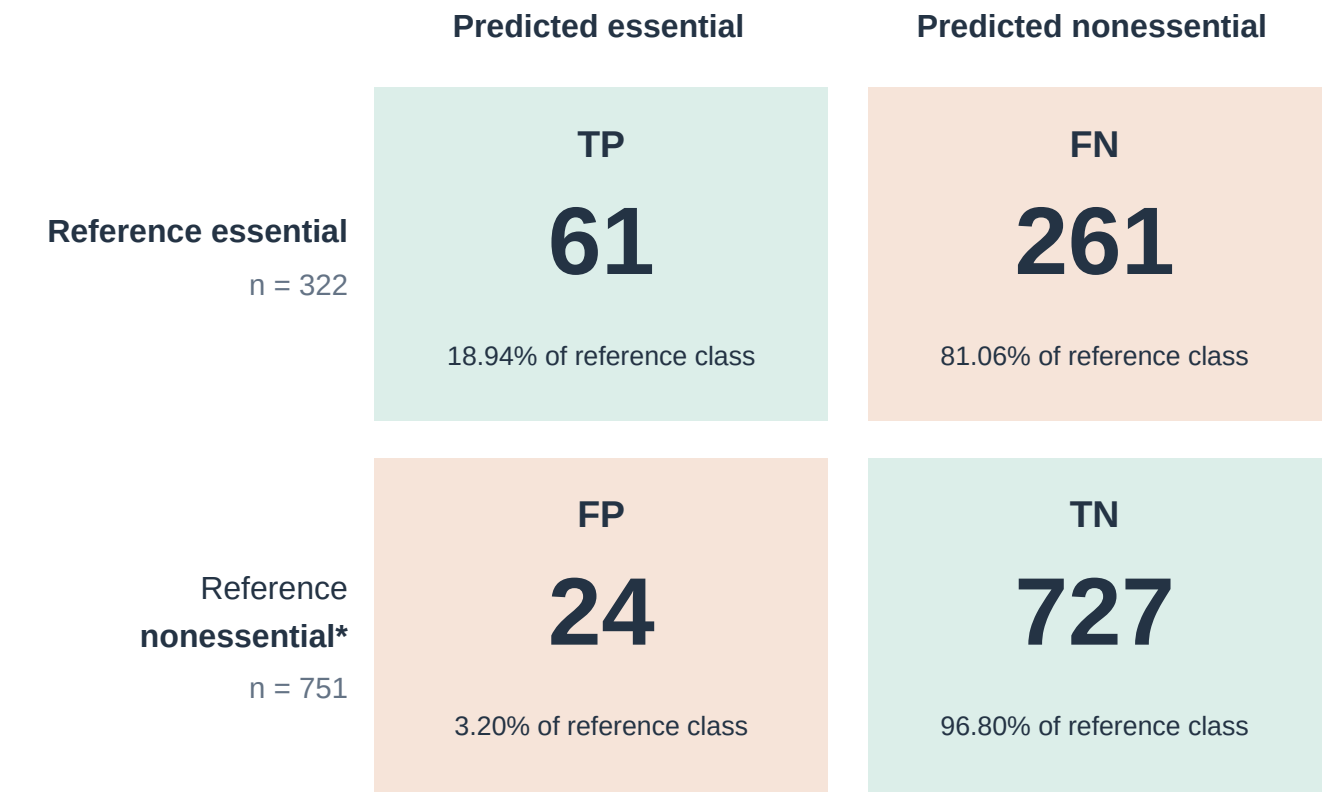


dFBA gene essentiality: E5 vs iYali21

24 h net biomass gain / WT < 15% | PO1f / CSM-Leu | Finite uracil | Initial biomass 0.05 gDW/L

Consensus-list membership benchmark | Euler step 0.1 h; classifications checked at 0.05 h and continuous limit

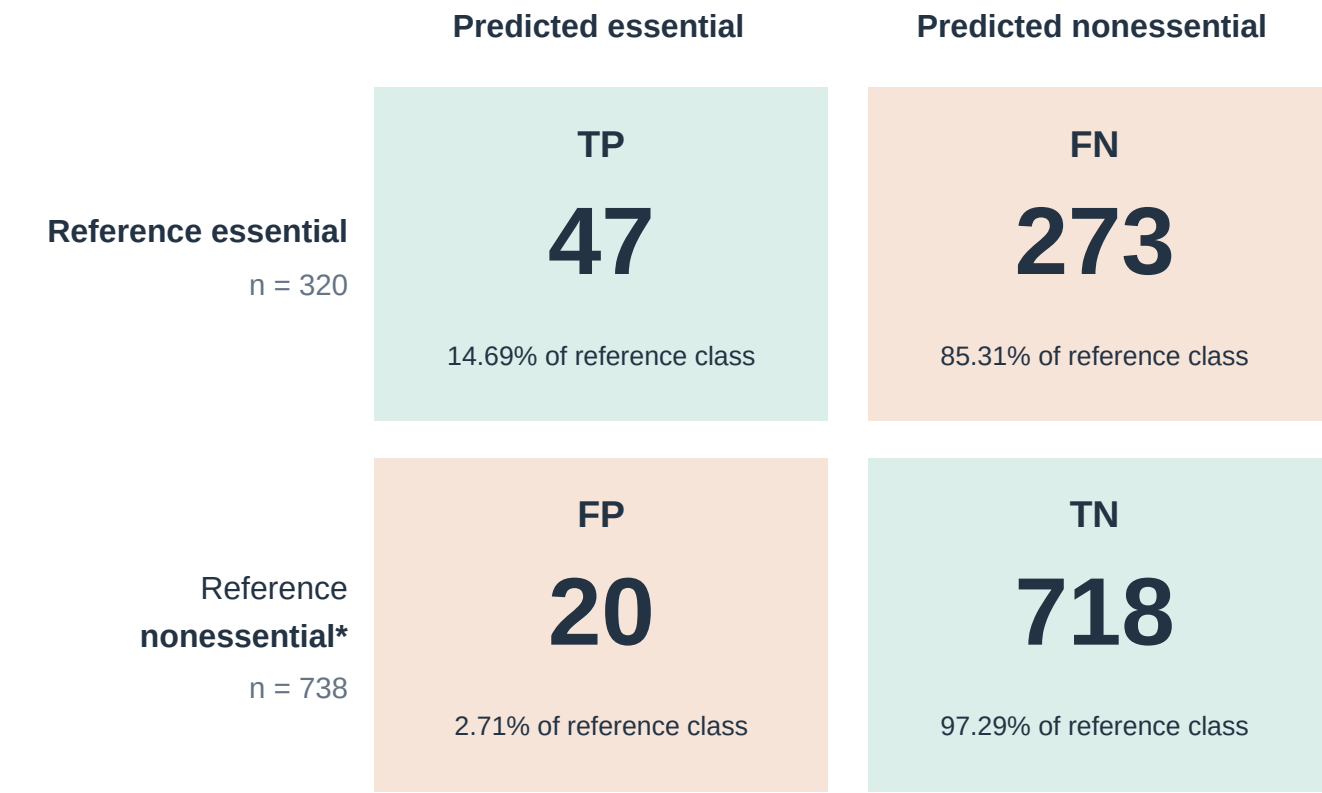
E5 | 1,073 model genes



Recall 18.94% | Precision 71.76% | Accuracy 73.44%

Unresolved: 0 | Step / continuous-limit class changes: 0 / 0

iYali21 | 1,058 model genes



Recall 14.69% | Precision 70.15% | Accuracy 72.31%

Unresolved: 0 | Step / continuous-limit class changes: 0 / 0

*Negative reference = model genes absent from the consensus essential list; experimental nonessentiality is not established.

No finite pool is depleted; availability caps remain inactive. KO endpoints use a verified constant-bound reduction of the dFBA equations.

This differs from the earlier static nonlimiting-uracil condition. Development/reference comparison; not independent validation. | 2026-10-05